

# MK Gold — Lead Integration & Automation Guide

Complete Technical Specification for mkgold.in & MK-GOLD CRM Real-Time Lead Ingestion

Platform: Next.js (Vercel) + Node.js/Express + MongoDB    Date: September 2026    Status: Ready for Implementation

## MK Gold: Complete Lead Integration & Automation Master Guide

This document provides a comprehensive technical breakdown of all public APIs on `mkgold.in`, the current lead flow into the `MK-GOLD CRM`, and the step-by-step implementation plan to automate real-time lead capture and telecaller assignment.

---

### Table of Contents

1. [System Architecture & Ecosystem Overview](#1-system-architecture--ecosystem-overview) 2. [Deep Dive: The 4 APIs Investigated](#2-deep-dive-the-4-apis-investigated) - [API 1 — Calculator Lead Form ( `POST /api/v1/leads` )](#api-1--calculator-lead-form) - [API 2 — Website Popup Lead Form ( `POST /api/v1/leads` )](#api-2--website-popup-lead-form) - [API 3 — Meta (Facebook & Instagram) Lead Ads Webhook ( `/api/v1/meta-leads` )](#api-3--meta-lead-ads-webhook) - [API 4 — Live Gold Valuation Calculator Engine ( `GET /api/v1/calculate` )](#api-4--live-gold-valuation-calculator-engine) 3. [Current Status vs. The Bottleneck (Live Database Findings)](#3-current-status-vs-the-bottleneck) 4. [Concepts Explained](#4-concepts-explained) - [What is a Webhook?](#what-is-a-webhook) - [What is an `x-api-key` & How to Generate It?](#what-is-an-x-api-key--how-to-generate-it) - [Hosting Confirmation: Vercel](#hosting-confirmation-vercel) 5. [End-to-End Implementation Plan](#5-end-to-end-implementation-plan) - [Part A: MK-GOLD CRM Backend (Webhook Receiver)](#part-a-mk-gold-crm-backend-webhook-receiver) - [Part B: mkgold.in Website (Webhook Forwarder)](#part-b-mkgoldin-website-webhook-forwarder) - [Part C: CRM Frontend (Telecaller Dashboard Badges & Filters)](#part-c-crm-frontend-telecaller-dashboard) 6. [Ready-to-Send Message for the `mkgold.in` Developer](#6-ready-to-send-message-for-the-mkgoldin-developer) 7. [Verification & Testing Checklist](#7-verification--testing-checklist)

---

# 1. System Architecture & Ecosystem Overview

## 1. TRAFFIC CHANNELS & TOUCHPOINTS (PUBLIC)

**Calculator Form**  
POST /api/v1/leads  
Gated modal on /sell-gold. Captures name, phone, gold weight & karat.  
`source: "calculator-gate"`

**Website Popup Modal**  
POST /api/v1/leads  
Auto-triggered after 2s or on CTA button. Captures customer details + notes.  
`source: "popup-lead-form"`

**Meta (FB & IG) Ads**  
POST /api/v1/meta-leads  
Instant Lead Forms from Facebook & Instagram targeted ad campaigns.  
`source: "meta-lead"`

**Live Valuation Engine**  
GET /api/v1/calculate  
Real-time MCX rate calculation (97.5% buying rate ±2%). No lead storage.  
`Rate Math Engine`

▼ Ingested by Website API Routes

## 2. WEBSITE APPLICATION LAYER (mkgold.in • Next.js on Vercel)

**Web API Handlers**  
/api/v1/leads • /api/v1/meta-leads  
Validates 10-digit Indian phone number • Saves local backup

→ Saves Local

**Website Database (SQL)**  
{ "success": true, "id": 711 }  
Internal website transaction backup

## SECURE AUTOMATED WEBHOOK (REAL-TIME BRIDGE)

Target: POST https://mkgold.tech/api/v1.0/public/leads • Auth: x-api-key: [PRE-SHARED-SECRET] • Speed: < 500ms

## 3. MK-GOLD CRM SYSTEM (Node.js/Express + MongoDB)

**STEP 1 Authentication & Ingestion**  
Validates x-api-key header. Rejects unauthorized traffic with 401.

→

**STEP 2 Normalize & Deduplicate**  
Maps phone → mobile. Checks for duplicate number on same day.

→

**STEP 3 Round-Robin Auto-Assign**  
Rotates lead evenly among all active telecallers:  
assignedTo: telecaller.\_id

→

**STEP 4 MongoDB Leads Store**  
Saves document into mk-gold.leads with leadSource: "marketing"

▼ Real-Time UI Delivery (< 2 Seconds)

## 4. TELECALLER & BRANCH CONVERSION QUEUE

**Telecaller Dashboard Queue**  
Lead pops up in assigned telecaller's queue with contact info, gold weight, karat, and city. Telecaller calls customer while interest is fresh.  
`Immediate Follow-Up`

**Branch Visit & Spot Payout**  
Telecaller schedules branch visit (Bangalore, Mysore, Mangalore, Davangere). In-branch non-destructive XRF test and payment in 30 mins.  
`Closed & Converted`

---

## 2. Deep Dive: The 4 APIs Investigated

### API 1 — Calculator Lead Form

**Endpoint:** POST `https://mkgold.in/api/v1/leads`

**Purpose:** Acts as a gatekeeper on [mkgold.in/sell-gold](https://www.mkgold.in/sell-gold#rate). The user must enter contact details before viewing their customized gold valuation.

**Payload Structure:**

```
{
  "name": "Prajwal Kumar",
  "phone": "9345004690",
  "city": "Bangalore",
  "gold_type": "jewellery",
  "weight_grams": 20,
  "purity_karat": 22,
  "source": "calculator-gate"
}
```

**Validation Rules:**

\* `name` and `phone` are strictly mandatory ( `400 {"error": "name and phone are required"}` ). \* `phone` must be a valid 10-digit Indian number matching `/^[6-9]\d{9}$/` ( `400 {"error": "Enter a valid 10-digit Indian mobile number"}` ).

**Post-Submit Action:** Returns `201 Created` with an auto-incrementing integer ID ( `{"success":true,"id":...}` ) and unlocks the rate calculator.

---

### API 2 — Website Popup Lead Form

**Endpoint:** POST `https://mkgold.in/api/v1/leads`

**Purpose:** Powers the promotional modal titled "Get a Free Gold Evaluation"\* that automatically slides in after **2 seconds** across homepage and landing pages.

**Payload Structure:**

```
{
  "name": "Padmaja",
  "phone": "9900119989",
  "city": "Bangalore",
  "gold_type": "jewellery",
  "weight_grams": 20,
  "purity_karat": 22,
  "notes": "Looking to release pledged gold from Muthoot",
  "source": "popup-lead-form"
}
```

**Difference from API 1:**

\* Contains an open `notes` field for custom remarks/queries. \* Tagged with `"source": "popup-lead-form"`. \* Triggers conversion tracking: `trackFormSubmit({ source: "popup" })`.

---

### API 3 — Meta (Facebook & Instagram) Lead Ads Webhook

**Endpoint:** `https://mkgold.in/api/v1/meta-leads`

**Purpose:** Captures leads automatically when a prospective customer fills out a lead form on a Facebook or Instagram ad.

#### Two Request Types:

1. **GET (Webhook Verification Handshake):** \* Called by Meta's developer servers using query parameters: `hub.mode=subscribe&hub.challenge=...&hub.verify_token=...`. \* **Current Issue:** Responding with `500 Server misconfigured` because the `META_VERIFY_TOKEN` environment variable has not been configured yet in Vercel.
2. **POST (Live Lead Event):** \* Meta sends a notification payload containing a `leadgen_id`. \* The server responds with `200 OK` (`{"status": "ok"}`) on valid Meta events, (`{"status": "ignored"}`) otherwise). \* The server uses `leadgen_id` to query Meta's Graph API and retrieve customer details.

---

### API 4 — Live Gold Valuation Calculator Engine

**Endpoint:** `GET https://mkgold.in/api/v1/calculate?weight=10&purity=22`

**Purpose:** Pure math calculation engine based on live MCX commodity rates.

**Does it capture leads? NO.** This endpoint does not accept name, phone, or any contact details.

**Live Test Output ( `200 OK` ):**

```
{
  "weight_grams": 10,
  "purity_karat": 22,
  "rate_per_gram": 13375,
  "mk_value": 130406,
  "estimate_low": 127798,
  "estimate_high": 133014,
  "mcx_rate_10g": 148000,
  "rate_24k": 14800,
  "rate_22k": 13375,
  "updated_at": "2026-09-05T04:57:46.279Z",
  "currency": "INR",
  "note": "Exact value confirmed at branch after certified XRF test."
}
```

#### Pricing Formula:

$$\text{MK Value} = \text{MCX Rate} \times \text{Weight (g)} \times \text{Purity} \times 97.5\%$$
 An estimate range (\$\pm 2\%\$) is provided to set customer expectations before non-destructive in-branch XRF testing.

---

## 3. Current Status vs. The Bottleneck

During our live test, we submitted a lead to `https://mkgold.in/api/v1/leads`:

The website returned `201 Created` with integer `id: 711`.

Direct query of the CRM database confirmed: **The lead did NOT appear in the CRM MongoDB.**

## What the Database Revealed:

1. In the CRM database, there are **7 Calculator leads** and **25+ Website Popup leads**. 2. **Every single one** has `createdBy: ObjectId("6a79c95d91ad3f799492e4b1")` (which is User `9740361350`, a `marketing` account). 3. Batches were imported on specific dates (Aug 13, Aug 21, Aug 25, Aug 31, Sep 2, Sep 4, Sep 5).

**WARNING: The Current Bottleneck:** The website and the CRM currently have **no live link**. The marketing staff is manually exporting CSVs from the website and importing them into the CRM every few days. This delays calling hot prospects by 24–48 hours.

## 4. Concepts Explained

### What is a Webhook?

A **Webhook** is an automated notification between two servers.

**Without a webhook (Manual):** You have to keep opening the website dashboard, checking for new leads, downloading Excel files, and uploading them to the CRM.

**With a webhook (Instant):** The exact millisecond a customer clicks "Submit" on `mkgold.in`, the website rings the CRM's digital doorbell, hand-delivering the lead.

### What is an `x-api-key` & How to Generate It?

An `x-api-key` is a shared secret password in the HTTP header that ensures only `mkgold.in` can push leads into the CRM, blocking unauthorized spam or bots.

#### How to generate one:

Run this in any terminal:

```
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

Example generated key:

```
mkgold_sec_e4b1a859f72c019958dcbe2da831ec5e921d782103f6f34e62a8435d21c43b91
```

## Hosting Confirmation: Vercel

`mkgold.in` is confirmed to be deployed on **Vercel** based on live HTTP headers:

`server: Vercel`

`x-vercel-id: bom1::bom1::...` (Mumbai region)

Media assets hosted on `pmcape3baplvglbo.public.blob.vercel-storage.com` (Vercel Blob Storage)

App framework: **Next.js**

## 5. End-to-End Implementation Plan

### Part A: MK-GOLD CRM Backend (Webhook Receiver)

#### 1. Add Secure Public Ingestion Route

In [ `backend/routes/public.js` ](file:///c:/Users/Mr.Green/Documents/Office/MKGOLD/MK-GOLD/backend/routes/public.js):

```
const leadPublicController = require("../controllers/public/lead");

function verifyWebhookKey(req, res, next) {
  const apiKey = req.headers["x-api-key"];
  if (!apiKey || apiKey !== process.env.CRM_WEBHOOK_KEY) {
    return res.status(401).json({ status: false, message: "Unauthorized: Invalid API Key" });
  }
  next();
}

router.post("/leads", verifyWebhookKey, leadPublicController.createWebLead);
```

## 2. Create Controller with Field Normalization & Auto-Assignment

Create [ `backend/controllers/public/lead.js` ](file:///c:/Users/Mr.Green/Documents/Office/MKGOLD/MK-GOLD/backend/controllers/public/lead.js):

```

const leadService = require("../services/lead");

async function createWebLead(req, res) {
  try {
    const { name, phone, city, gold_type, weight_grams, purity_karat, notes, source } = req.body;

    if (!name || !phone) {
      return res.status(400).json({ status: false, message: "name and phone are required" });
    }

    const cleanMobile = String(phone).replace(/\s+/g, "");

    // Map source labels
    let sourceLabel = "Website Lead";
    if (source === "calculator-gate") sourceLabel = "Calculator Lead Form";
    else if (source === "popup-lead-form") sourceLabel = "Website Popup";
    else if (source === "meta-lead") sourceLabel = "Meta Lead Ads";

    const remarksParts = [];
    if (notes) remarksParts.push(`Notes: ${notes}`);
    if (gold_type) remarksParts.push(`Type: ${gold_type}`);
    if (purity_karat) remarksParts.push(`Purity: ${purity_karat}K`);

    const leadPayload = {
      name: name.trim(),
      mobile: cleanMobile,
      city: city || "",
      place: city || "",
      source: sourceLabel,
      leadSource: "marketing",
      category: "gold",
      weight: parseFloat(weight_grams) || 0,
      remarks: remarksParts.join(" | "),
      date: new Date()
    };

    // Automatically assigns to an active telecaller via round-robin
    const createdLead = await leadService.create(leadPayload);

    return res.status(201).json({
      status: true,
      message: "Lead created and assigned successfully",
      data: { id: createdLead._id, assignedTo: createdLead.assignedTo }
    });
  } catch (err) {
    // If already exists today, return clean response without breaking
    return res.status(200).json({ status: false, message: err.message });
  }
}

module.exports = { createWebLead };

```

### 3. Store the Secret Key in CRM .env

In [ backend/.env ](file:///c:/Users/Mr.Green/Documents/Office/MKGOLD/MK-GOLD/backend/.env):

```
CRM_WEBHOOK_KEY="mkgold_sec_e4b1a859f72c019958dcbe2da831ec5e921d782103f6f34e62a8435d21c43b91"
```

## Part B: mkgold.in Website (Webhook Forwarder)

In the `mkgold.in` Next.js repository (under `/api/v1/leads` or `/api/leads`):

```
// Inside POST handler of /api/v1/leads (after saving to local DB):
try {
  fetch("https://mkgold.tech/api/v1.0/public/leads", {
    method: "POST",
    headers: {
      "Content-Type": "application/json",
      "x-api-key": process.env.CRM_WEBHOOK_KEY
    },
    body: JSON.stringify({
      name: body.name,
      phone: body.phone,
      city: body.city,
      gold_type: body.gold_type,
      weight_grams: body.weight_grams,
      purity_karat: body.purity_karat,
      notes: body.notes,
      source: body.source // "calculator-gate" or "popup-lead-form"
    })
  }).catch((e) => console.error("CRM Webhook push failed:", e));
} catch (err) {
  console.error("Webhook trigger error:", err);
}
```

In **Vercel Settings**  $\rightarrow$  **Environment Variables**:

Add `CRM_WEBHOOK_KEY`

Add `META_VERIFY_TOKEN`

## Part C: CRM Frontend (Telecaller Dashboard)

1. In `[frontend/src/sections/@dashboard/lead/LeadFilterSidebar.js]`

(file:///c:/Users/Mr.Green/Documents/Office/MKGOLD/MK-

GOLD/frontend/src/sections/@dashboard/lead/LeadFilterSidebar.js), ensure telecallers can filter by:

\* `Calculator Lead Form`

\* `Website Popup`

\* `Meta Lead Ads`

2. In `[frontend/src/pages/marketing/Leads.js]` (file:///c:/Users/Mr.Green/Documents/Office/MKGOLD/MK-

GOLD/frontend/src/pages/marketing/Leads.js), display a distinctive chip badge next to web leads so telecallers know the exact context when they initiate the call.

## 6. Ready-to-Send Message for the `mkgold.in` Developer

Copy and send this to the website developer:



## Subject: Real-Time Lead Webhook Integration to MK-GOLD CRM

> Hi,

> We want leads submitted on **mkgold.in** (both from the **Calculator Form** and **Popup Form**) as well as **Meta Leads** to automatically push into our CRM so telecallers receive them instantly.

> Could you please make these 2 updates on the website?

> ### 1. Forward Form Submissions to CRM Webhook

In your `POST /api/v1/leads` (and `/api/leads`) handler, right after saving to your local database, please forward the lead to:

**URL:** `https://mkgold.tech/api/v1.0/public/Leads` (or production CRM URL)\*

\* **Method:** `POST`

\* **Headers:**

\* `Content-Type: application/json`

\* `x-api-key: [PASTE_GENERATED_API_KEY_HERE]`

\* **Payload to forward:**

```
> {
>   "name": name,
>   "phone": phone,
>   "city": city,
>   "gold_type": gold_type,
>   "weight_grams": weight_grams,
>   "purity_karat": purity_karat,
>   "notes": notes,
>   "source": source
> }
```

> ### 2. Fix Meta Leads Webhook ( `/api/v1/meta-leads` )

`GET /api/v1/meta-leads` currently returns `500 Server misconfigured` .

\* Please set `META_VERIFY_TOKEN` in the **Vercel Project Environment Variables** so Meta webhook verification can

succeed.

\* When a new lead is received from Meta, forward it to the same CRM webhook with `"source": "meta-lead"`.

> Please let us know once this is deployed so we can run an end-to-end test.

---

7. Verification & Testing Checklist

- ☐ **API Key Check:** Request to `/api/v1.0/public/leads` without `x-api-key` returns `401 Unauthorized`.
- ☐ **Validation Check:** Submitting missing name/phone returns `400 Bad Request`.
- ☐ **Auto-Assignment Check:** Submitting a valid lead saves into MongoDB `leads` and populates `assignedTo` with an active telecaller.
- ☐ **Deduplication Check:** Submitting the same phone number twice on the same day handles gracefully without creating duplicates.
- ☐ **End-to-End Live Check:** Fill in the calculator form on `mkgold.in` and confirm the lead appears on the telecaller's screen in under 2 seconds.